

B.Tech Project Phase I Report
on
NEXRAG: RAG CHATBOT FOR EDUCATION

*Submitted in partial fulfilment for the award of the Degree of
Bachelor of Technology in Computer Science and Engineering*

Submitted by
ADNAN SAIF (Reg.No: MEK22CS003)
AJAS N (Reg.No: MEK22CS006)
AJMAL AHAMMED KABEER (Reg.No: MEK22CS007)
AL AMEEN (Reg.No: MEK22CS009)

Under the guidance of

Prof.MUNEERA



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
MES INSTITUTE OF TECHNOLOGY AND MANAGEMENT,
CHATHANNOOR, KOLLAM

September 2025

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
MES INSTITUTE OF TECHNOLOGY AND MANAGEMENT, KOLLAM



CERTIFICATE

This is to certify that the project entitled "***NEXRAG: RAG CHATBOT FOR EDUCATION***" is a Bonafide record of the project done by **ADNAN SAIF** (Reg.No:MEK22CS003), **AJAS N** (Reg.No:MEK22CS006), **AJMAL AHAMMED KABEER** (Reg.No:MEK20CS007) and **AL AMEEN** (Reg.No:MEK20CS009) of Seventh semester, under my supervision and guidance, in partial fulfillment for the award of Degree of Bachelor of Technology in Computer Science and Engineering from the A P J Abdul Kalam Technological University for the year 2025.

Prof.Muneera
Assistant Professor
(Guide)
Dept. of CSE

Prof.Sajinsha N
Assistant Professor
(Project Co-Ordinator)
Dept. of CSE (AI)

Prof.Saheer H
Associate Professor & HoD
Dept. of CSE

Place: Kollam

Date:

ACKNOWLEDGEMENT

First and foremost, we wish to place on records our ardent and earnest gratitude to our project guide **Prof.Muneera**, *Assistant Professor, Dept. Of Computer Science and Engineering*. Her tutelage and guidance was the leading factor in translating our efforts to fruition. Her prudent and perceptive vision has shown on our trail to triumph.

We would like to express our sincere thanks to our respected *Principal in charge*, **Prof.Rafi A** for facilities provided for doing this project.

We are extremely happy to mention a great word of gratitude to **Prof.Saheer H**, *Head of the Department, Computer Science and Engineering* for providing us with all facilities for the completion of this work.

Finally, yet importantly, we would like to express our gratitude to our project coordinator **Prof.Sajinsha N**, *Assistant Professor, Dept. Of Computer Science and Engineering (Artificial Intelligence)* for his valuable assistance during the course of the project.

We also extend our gratefulness to all the staff members in the department. We also thank all our friends and well-wishers who greatly helped us in our endeavor.

ADNAN SAIF (Reg. No: MEK22CS003)

AJAS.N (Reg. No: MEK22CS006)

AJMAL AHAMMED KABEER (Reg. No: MEK22CS007)

AL AMEEN (Reg. No: MEK22CS009)

ABSTRACT

Students often face difficulties in finding accurate and timely answers to their academic and personal queries, which may include course details, study resources, exam preparation, scholarships, and career guidance. Traditional methods such as browsing university websites or depending on administrative staff are often time-consuming and overwhelming. To address this issue, NexRag is proposed as a student-focused chatbot powered by Retrieval-Augmented Generation (RAG). The system is designed to efficiently process student-specific queries and generate precise, context-aware answers by integrating three core components: knowledge ingestion from academic and student-related resources, vector-based retrieval using embeddings for fast and accurate search, and large language model-driven response generation for personalized interaction. This combination ensures that students receive relevant and reliable information instantly. NexRag aims to reduce the dependency on manual information sources and streamline the student support process. By offering instant, reliable, and student-centered assistance, the chatbot enhances the overall learning experience and helps students access the right guidance at the right time. This innovative solution holds the potential to foster academic success while enabling smoother navigation through various aspects of university life.